

Sujith K

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PROFILE

Backend & AI Engineer specializing in multi-agent systems, LLM orchestration, and distributed microservice architectures. Designed modular agent frameworks, centralized LLM abstraction layers, and RAG-enabled services using LangGraph, FastAPI, and vector indexing. Experienced in building scalable, protocol-driven AI platforms with real-time execution tracking and production-grade deployment practices.

SKILLS

Backend & APIs: Python | FastAPI | REST APIs | WebSockets | OAuth 2.0

LLM & Agents: LangGraph | LangChain | Multi-Agent Orchestration | Retrieval-Augmented Generation (RAG) | Prompt Architecture | LLM Abstraction Layers | Agent Orchestration | MCP Servers

Architecture & Systems: Microservices Architecture | Hub-and-Spoke Systems | Distributed Services | Custom Protocol Design | JSON-RPC

Data & Indexing: FAISS | Vector Search | Structured LLM Output Parsing

Infrastructure: Docker | Uvicorn | Git | Environment Configuration Management

PROFESSIONAL EXPERIENCE

Apexon, Backend Engineer – Agentic Systems

06/2024 – Present | Bengaluru

- Designed and implemented a centralized LLM adapter layer supporting multiple LLM providers, enabling seamless provider switching across services.
- Re-architected backend services into modular multi-agent workflows using LangGraph (analysis, planning, transformation, review, reporting).
- Built automated code migration pipelines (COBOL → Java, C++ → Java, Java upgrades) with structured planning, generation, validation, and testcase agents.
- Developed node-based workflow orchestration engine with deterministic prompt templates and structured state handling.
- Implemented WebSocket-based real-time execution tracking for long-running agent workflows.
- Hardened backend reliability with structured error contracts, schema-based LLM outputs, persona validation, and robust parsing.
- Improved deployment stability through Docker/Uvicorn tuning and standardized environment management.
- Led API refactoring and versioning initiatives to reduce technical debt and improve service maintainability.

PROJECTS

A2A Travel Planner – Multi-Agent LLM Platform

Tech: Python, FastAPI, LangGraph, CrewAI, React (TypeScript), OpenAI GPT-4o, A2A Protocol

- Designed and implemented a full-stack multi-agent travel planning system demonstrating Agent-to-Agent (A2A) interoperability between LangGraph and CrewAI frameworks.
- Built a FastAPI-based orchestrator that parses natural language requests and dynamically dispatches tasks to domain-specific agents (Flights, Hotels, Cars).
- Implemented modular, independently deployable agent services exposing standardized agent cards and JSON-RPC communication via python-a2a SDK.
- Integrated OpenAI GPT-4o for intent parsing, structured reasoning, and intelligent travel plan generation.
- Designed microservice architecture with clear separation of concerns, environment isolation, and reproducible local orchestration.

MCP_OAuth – Modular OAuth Authentication & Token Platform

Tech: Python, FastAPI, OAuth 2.0, JavaScript, REST APIs

- Designed and implemented a modular OAuth 2.0 authentication system supporting provider-based login (e.g., GitHub) with secure token exchange and validation.
- Built RESTful API endpoints for login, token issuance, protected resource access, and session management.
- Architected clean separation between authentication providers, protocol handlers, and service layers for extensibility and maintainability.
- Implemented secure token handling practices including validation, storage abstraction, and access control enforcement.
- Structured the project using reusable modules to enable future multi-provider support and scalability.

MCP Platform – Modular Context Protocol & Agent Hub

Tech: Python, FastAPI, FAISS, REST APIs, Custom Protocols, HTML/JavaScript

- Architected a hub-and-spoke microservice platform for orchestrating AI agents, RAG services, and external data providers.
- Developed a central coordination hub responsible for agent routing, configuration management, and database interactions.
- Implemented modular backend agents for task execution, service communication, and data processing.
- Built specialized microservices including a Retrieval-Augmented Generation (RAG) server with FAISS-based vector indexing and a weather service API.
- Designed REST and custom protocol interfaces enabling flexible client-server communication.
- Created Python client libraries to standardize service interaction and simplify integration of new agents or data sources.
- Structured the system for extensibility, allowing new services to be added with minimal architectural changes.

EDUCATION

Bachelor of Engineering in Electronics & Communication Engineering

2019 – 2023 | Mangaluru

Canara Engineering College — Visvesvaraya Technological University

CERTIFICATES

LangChain – Develop AI Agents with LangChain & LangGraph

Udemy — Eden Marco

Completed: 2025